

Jay Patil

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Education

Rutgers University, New Jersey – *Master of Science in Computer Science*

Sept 2022 - May 2024

COURSES: Algorithms, Machine Learning, Distributed Systems, OS, DBMS

CGPA: 3.71/4.00

Pune Institute of Computer Technology, Pune – *Bachelor of Engineering in Computer Engineering*

Aug 2016 - June 2020

COURSES: Advanced Data Structures, Computer Networks, Computer Architecture

CGPA: 8.97/10.00

Experience

Antonina Mitrofanova Lab – Research Assistant, Software Developer

Sept 2022 – May 2024

- Designed and developed a **user-engaging and cross-browser compatible** web application for data analysis and rendering of complex human gene-splicing networks consisting of more than 2k nodes using Angular, and Cytoscape.js.
- Implemented **data feeds** from sources like NCBI ensuring seamless data integration and real-time updates.
- Conducted **Principal Component Analysis** on human gene dataset and reduced dimensionality to improve compute time of training and eliminated unnecessary correlated variable.

Veritas Technologies LLC – Software Engineer, Full Stack

Jul 2020 – Jul 2022

- Worked in a **Scrum-based agile** environment and developed intuitive user interfaces using Angular to facilitate the data discovery of more than **1 billion** messages and emails from major data sources like Teams, Outlook, Slack, etc.
- Took the ownership to update old code base to modern development standards making it **more secure** by implementing API authorization in .NET Core and **reducing loading time** of the application by **46%** by implementing lazy loading of pages.
- Leveraged Elasticsearch to implement fuzzy search on more than a billion MS Teams messages to retrieve relevant data.
- Developed a web service to integrate and connect One Drive services from Merge1 to Veritas' internal platform.
- Implemented Azure Storage REST APIs using C# in .NET to retrieve data from Blob storage to support the email export service.
- Optimized stored procedures in MSSQL by using **indexed columns** in queries and writing modular code.
- Coordinated efficient large-scale features deployments using Terraform and Puppet and managed CI/CD using TFS.

Projects

Text-to-Video Generation through Advanced Diffusion Models (GenAI) | Python, Linux, TensorFlow

- Spearheaded a pioneering text-to-video (T2V) generation project utilizing the "Tune-A-Video" methodology, focused on **one-shot tuning** of image diffusion models to elevate the quality and contextual accuracy of video output across diverse data domains. Utilized TensorFlow and PyTorch frameworks to meticulously implement and fine-tune the model.

Attention Mechanisms in Image Captioning | Python, Linux, PyTorch, NumPy, Pandas

- Led a comprehensive comparative analysis of RATT, XLAN, and Dual Attention models in Computer Vision. Utilized TensorFlow and PyTorch frameworks to meticulously implement and fine-tune the models. Employed the METEOR metric to assess the quality of complex texts, providing nuanced insights into the strengths and potential areas for enhancement of each model.

Distributed Data Store Optimization | Linux, C, dqlite, Raft

- Implemented a query batching and pipelining optimizations to a distributed highly available key value store(dqlite) built using sqlite and raft consensus protocol (open-source canonical implementation) achieving **10% reduction in response time**. Created benchmarks to measure performance of different scenarios like node failure, membership changes, network partition.

Skills

Languages: Python3, C#, Java, JavaScript, SQL, C/C++, HTML/CSS, TypeScript

Web Technologies: SpringBoot, NodeJS, React, Express, Angular, Flask, Web3, O'Auth, FastAPI

Tools and Frameworks: Git, Azure, Docker, MSSQL, MongoDB, Kubernetes, ASP .NET, Elasticsearch, LangChain, Kafka

Awards & Recognition

- Award for 'Team Collaboration and Inclusion' for collaboration with teams across the globe at Veritas.